Debarati Chakrabarti

New York, NY | (332) 250-0392 | cdebarati10@gmail.com | linkedin.com/in/debaratichakrabarti | Website

Education

Columbia University — New York, NY
Master of Science, Computer Science — *Aug 2026 - Dec 2027*

Meghnad Saha Institute Of Technology — Kolkata, India
Bachelor of Technology, Electronics and Communications Engineering CGPA - 8.86/10 — *Jul 2019 - Jul 2023*

Experience

Columbia Business School — New York, NY
Machine Learning Engineer — *Sep 2026 - Dec 2026*

- Part of the Columbia Build Lab Fall 2026 cohort, building ML pipelines for a venture's core technology and product.

The Bank of New York — Pune, India
Associate - Full-Stack Engineer — *Mar 2025 - Aug 2026*

- Led problem statement formulation and Retrieval-Augmented Generation **(RAG) based LLM** application development in Asset Servicing's **AI hackathon**, securing **1st position** in Round 1 and **Runner-Up** in Round 2 by automating e-form processing for key client APIs (**BlackRock, Primerica**).
- Collaborated with business teams to expand the solution into **2026 AI Initiatives**, reduced costs by 3.5 FTE hours per API to save **$40K/year** and projecting **$700K+** annual savings via platform-wide automation of 100s of e-forms.
- Migrated a major legacy application from **ExtJS to Angular**, implementing 20+ UI screens with complex business rules, performed UI-level caching and reduced page load time by **22%**.
- Implemented a distributed caching layer using **Hazelcast**, cutting p95 API response times by **40%** and enabling the backend to sustain **2x** higher concurrent throughput.
- Mentored a team of 3 summer interns and 2 new grad analysts, led **Agile** sprints, and coordinated cross-functional delivery with operations and product teams.

Analyst - Software Developer — *Jul 2023 - Mar 2025*

- Migrated Nuance, a vendor **Groovy** application used for investor and audit operations to a full-stack web app leveraging **Java, Spring Boot** and **Angular**.
- Owned complete front-end development of Nuance using **Angular, Typescript** implementing reusable components and modular architecture, decreasing code duplication by **42%**.
- Re-engineered enterprise REST services in **Spring Boot with Oracle DB**, optimizing data access via query refactoring and indexing to slash execution time by **55%** and lowered **peak CPU usage**.
- Designed and implemented batch jobs to replace legacy workflows; optimized scheduling and chunk processing, decreasing batch execution time by **17%**.
- Integrated **Amazon S3** for scalable audio storage, automating secure file access across client workflows to reliably handle high volumes of audio files.
- Built an automated **Python** solution integrating OCR to extract unstructured PDF data and process complex business logic from multi-sheet Excel files, reducing manual effort in investment operations.
- Cleaned up legacy **Java/Spring Boot** code, reaching **85%** code coverage and fixing critical SonarQube vulnerabilities.

Coding Ninjas
Teaching Assistant - C++ Data Structures and Algorithms — *Dec 2022 - Apr 2023*

- Mentored **500+** students in **C++ DSA** and **OOP** through doubt and debugging sessions, rated **4.96/5.0**.

Projects

Track My Ferry - Real-Time NYC Ferry Arrival Prediction — GitHub

- Built a live NYC Ferry tracker that works out which trip each boat is running and predicts arrival time ranges and crowding.
- Streamed live AIS, **GTFS-realtime** and NOAA tide data through **Kafka** into **PostgreSQL/TimescaleDB**, matching boats to scheduled trips with **99.4%** accuracy, and served a live map over **FastAPI WebSockets** and **React + MapLibre**.
- **Quantile GBMs** with **conformal calibration**: **48s** mean arrival error vs 92s baseline, **81%** interval coverage (simulated).

Sentiment Analysis of Bengali-English-Hindi Code-Mixed Text: Classical vs Transformer Models
(Primary Author, IEEE SIGNASS 2026)

- Conducted comparative evaluation of classical ML models **(SVM F1 0.677, Naive Bayes F1 0.65)** and transformer-based multilingual **PLMs (mBERT F1 0.75, XLM-RoBERTa F1 0.73)**, showing a **10%+ gain** with transformers.
- Demonstrated impact of multilingual pretraining and cross-lingual alignment for **9.5K trilingual code-mixed dataset**, establishing a rare trilingual benchmark dataset for Indian languages and enabling reproducible evaluation for future research.

Skills

- **Languages**: **Java, C++, Python, SQL, TypeScript**, JavaScript, HTML, CSS
- **Frameworks & Libraries**: Spring Boot, Hibernate/JPA, Angular, ExtJS, React, Redux, Node.js, FastAPI, JUnit, Jest
- **AI/ML**: LLMs, RAG, Transformers (mBERT, XLM-RoBERTa), scikit-learn, TensorFlow, Pandas, NumPy, OCR
- **Databases & Streaming**: Oracle DB, PostgreSQL, TimescaleDB, Hazelcast, Kafka, WebSockets
- **Cloud & Tools**: AWS (S3), Docker, Git, GitHub Actions, CI/CD, SonarQube, REST APIs, Microservices, Agile/Scrum
- **Coursework**: DSA, OOP, Distributed Systems, Operating Systems, DBMS, Computer Networks, Software Engineering, ML, AI